

Chapter 2

Interest Rates

2.1 Types of rates

◇ An interest rate (IR, hereafter) in a particular situation defines the amount of money a borrower promises to pay the lender.

- For any given currency, many different types of IRs are regularly quoted (e.g., mortgage rates, deposit rates, and CD rates).
- The IR applicable in a situation depends on the *credit risk*: the higher the credit risk, the higher the IR that is promised by the borrower.

◇ Treasury rates are the rates an investor earns on Treasury bills (or T-bills) and Treasury bonds (or T-bonds). These are instruments used by a government to borrow in its won currency (e.g., Korean Treasury rates)

- Treasury rates are totally *risk-free* rates in the sense that an investor who buys a T-bill or T-bond is certain that interest and principal payments will be made as promised.
- Derivative traders do not usually use Treasury rates as risk-free rates. Instead, they use LIBOR rates due to tax and regulatory issues.

◇ A LIBOR (London Interbank Offered Rate) quoted by a particular bank is the IR at which the bank is prepared to make a large wholesale deposit with other banks.

- Typically the deposit has a AA credit rating (the second best credit rating scored by S&P).
 - LIBOR rates are not totally free of credit risk, but close to risk-free.
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- CD (Certificate of Deposit) rates at which the large banks issue them in the money market.
- ◇ A Repo (repurchase agreement, RP) rate is the rate of interest defined as the difference between the price at which the securities are sold and the price at which they are repurchased.
- The most common type of repo is an *overnight repo*, in which the agreement is renegotiated each day. However, longer-term arrangements, known as *term repos*, are sometimes used.

2.2 Measuring interest rates: compounding

<Effect of the compounding frequency>

Compounding frequency	Value of \$100 in one year at 10%
Annual (m=1)	110.00
Semiannual (m=2)	110.25
Quarterly (m=4)	110.38
Monthly (m=12)	110.47
Weekly (m=52)	110.51
Daily (m=365)	110.52

Figure 2.1: Effect of the compounding frequency on the value of \$100 at the end of 1 year when the interest rate is 10% per annum.

- ◇ Conversion formulas: Define R_c as continuously compounded IR and R_m as the same IR with compounding m times per year, then

$$R_c = m \log \left(1 + \frac{R_m}{m} \right), \text{ or equivalently, } R_m = m(e^{R_c/m} - 1).$$

- IRs used in option pricing are nearly always expressed with continuous compounding.

2.3 Zero rates

- ◇ A T -year zero rate (or spot rate, or zero, or zero-coupon IR) is the IR earned on an investment that provides a payoff only at time T .
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Example 2.3.1. (Zero rates with different maturities)

Consider the situation where Treasury zero rates, measured with continuous compounding, are as in the following Table.

Maturity (years)	Zero rate (cont. comp.)
0.5	5.0
1.0	5.8
1.5	6.4
2.0	6.8

Figure 2.2: Treasury zero rates with different maturities

2.4 Bond pricing

◇ To calculate the theoretical (cash) price of a bond we discount each cash flow at the appropriate zero rate. Assume the notional principal is \$100.

- In Example 2.3.1, the theoretical price of a two-year bond providing a 6% coupon semiannually is

$$3e^{-0.05 \times 0.5} + 3e^{-0.058 \times 1.0} + 3e^{-0.064 \times 1.5} + 103e^{-0.068 \times 2.0} = 98.39.$$

◇ The *bond yield* is the discount rate that makes the present value of the cash flows on the bond equal to the market price of the bond.

- In Example 2.3.1, Suppose that the market price of the bond with face value of \$100 equals its theoretical price of \$98.39. The bond yield (continuously compounded) is given by solving

$$3e^{-y \times 0.5} + 3e^{-y \times 1.0} + 3e^{-y \times 1.5} + 103e^{-y \times 2.0} = 98.39,$$

to get $y = 0.0676$ or 6.76%.

◇ The *par yield* for a certain maturity is the coupon rate (not coupon payment) that causes the bond price to equal its face value.

- In Example 2.3.1, we solve

$$\frac{c}{2}e^{-0.05 \times 0.5} + \frac{c}{2}e^{-0.058 \times 1.0} + \frac{c}{2}e^{-0.064 \times 1.5} + \left(100 + \frac{c}{2}\right)e^{-0.068 \times 2.0} = 100,$$

to get $c/100 = 6.87\%$ with semiannual compounding.

- In general if m is the number of coupon payments per year, d is the present value of \$1 received at maturity and A is the present value of an annuity of \$1 on each coupon date

$$\text{par yield} = \frac{(1 - d)m}{A}.$$

(In Example 2.2, $m = 2$, $d = 0.87284$, and $A = 3.70027$)

2.5 Determining Treasury zero rates

◇ The *bootstrap method* is one way to determine Treasury zero rates from T-bills and coupon-bearing bonds.

Example 2.5.1. (Bootstrap method)

Consider the following data for the bootstrap method.

Bond Principal	Time to Maturity (yrs)	Coupon per year (\$)*	Bond price (\$)
100	0.25	0	97.5
100	0.50	0	94.9
100	1.00	0	90.0
100	1.50	8	96.0
100	2.00	12	101.6

Figure 2.3: *Half the stated coupon is assumed to be paid every six months.

- An amount 2.5 can be earned on 97.5 during 3 months.
- Because $100 = 97.5e^{0.10127 \times 0.25}$, the 3-month rate is 10.127% with continuous compounding.
- Similarly the 6 month and 1-year rates are 10.469% and 10.536% with continuous compounding.
- To calculate the 1.5-year rate we solve

$$4e^{-0.10469 \times 0.5} + 4e^{-0.10536 \times 1.0} + 104e^{-R \times 1.5} = 96,$$

to get $R = 0.10681$, or 10.681%.

- Similarly, the 2-year rate is 10.808%.

Then, we can draw the zero curve calculated from the data:

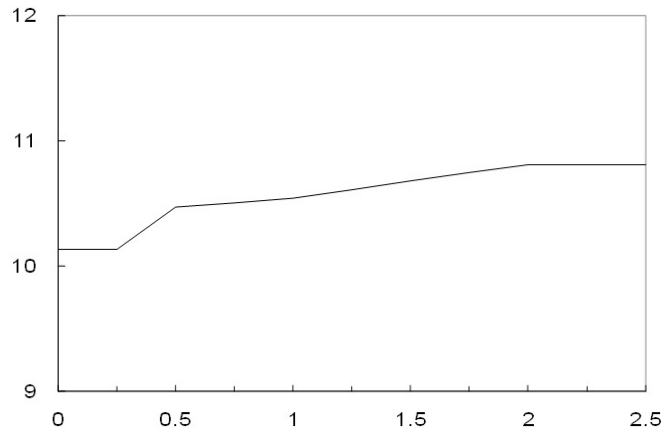


Figure 2.4: Zero rate as a function with respect to maturity.

2.6 Forward rates

◇ *Forward (interest) rates* are the rates of interest implied by current zero rates for periods of time in the future.

- Suppose that the zero rates for time periods T_1 and T_2 are R_1 and R_2 with both rates continuously compounded. The forward rate f_{12} for the period between times T_1 and T_2 is

$$f_{12} = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}.$$

This formula is only approximately true when rates are not expressed with continuous compounding.

Example 2.6.1. (Forward rates)

From the formula, we can get the forward rates in Figure 2.5.

◇ The *instantaneous* forward rate for a maturity T is the forward rate that applies for a very short time period starting at T . It is

$$R + T \frac{\partial R}{\partial T}$$

where R is the T -year IR.

◇ For an upward sloping yield curve, we easily find that (forward rate) > (zero rate) > (par yield). The opposite is true for a downward sloping yield curve. (**Why? Prove it for simple examples.**)

Year (n)	Zero rate for n -year investment (% per annum)	Forward rate for n th year (% per annum)
1	3.0	
2	4.0	5.0
3	4.6	5.8
4	5.0	6.2
5	5.5	6.5

Figure 2.5: Application for the formula

2.7 Forward rate agreements

◇ A forward rate agreement (FRA) is an over-the-counter (OTC) agreement that a certain rate will apply to a certain principal during a certain future time period.

- An FRA is equivalent to an agreement where interest at a predetermined rate, R_K , is exchanged for interest at the market rate.
- An FRA can be valued by assuming that the forward CD rate, R_F , is certain to be realized.
- This means that the value of an FRA is the present value of the difference between the interest that would be paid at interest at rate R_F and the interest that would be paid at rate R_K .

◇ Valuation formulas

- If the period to which an FRA applies lasts from T_1 to T_2 , we assume that R_F and R_K are expressed with a compounding frequency corresponding to the length of the period between T_1 and T_2 .
- With an interest rate of R_K , the interest cash flow is $R_K(T_2 - T_1)$ at time T_2 and, with an interest rate of R_F , it is $R_F(T_2 - T_1)$ at time T_2 .
- When the rate R_K will be received on a principal of L , the value of the FRA is the present value of $L(R_K - R_F)(T_2 - T_1)$, received at time T_2 .
- When the rate R_F will be received on a principal of L the value of the FRA is the present value of $L(R_F - R_K)(T_2 - T_1)$, received at time T_2 .

Example 2.7.1. (FRA)

An FRA entered into some time ago ensures that a company will receive 4%

(semiannually) on \$100 million for six months starting in 1 year. The forward CD for the period is 5% (semiannually) and the 1.5-year rate is 4.5% with continuous compounding. Then, the value of the FRA (in \$ millions) is

$$100 \times (0.04 - 0.05) \times 0.5 \times e^{-0.045 \times 1.5} = -0.467.$$

If the six-month interest rate in one year turns out to be 5.5% (s.a.) there will be a payoff (in \$ millions) of

$$100 \times (0.04 - 0.055) \times 0.5 = -0.75$$

in 1.5 years. The transaction might be settled at the 1-year point for an equivalent payoff of $(-0.75/1.0275) = -0.730$.

2.9 Duration

◇ The (Macaulay) *duration* of a bond that provides cash flow c_i at time t_i ($i = 1, 2, \dots, n$ and $t_1 < t_2 < \dots < t_n$) is

$$D = \sum_{i=1}^n t_i \frac{c_i e^{-yt_i}}{B}$$

where B is its price, $\sum_{i=1}^n c_i e^{-yt_i}$ and y is its yield (continuously compounded).

- Duration is important because it leads to the following key relationship between the change in the yield on the bond and the change in its price:

$$\frac{\Delta B}{B} = -D \Delta y.$$

◇ The *modified duration* of a bond is defined as

$$D^* = \frac{D}{1 + y/m}.$$

- When the yield y is expressed with compounding m times per year,

$$\frac{\Delta B}{B} = -\frac{D \Delta y}{1 + y/m} = -D^* \Delta y.$$

◇ The *dollar duration*, D^{**} , of a bond is defined as the product of modified duration and bond price, so that

$$\Delta B = -D^{**} \Delta y.$$

◇ The duration for a bond portfolio is the weighted average duration of the bonds in the portfolio with weights proportional to prices.

- The key duration relationship for a bond portfolio describes the effect of small parallel shifts in the yield curve.
- By choosing a portfolio so that the duration of assets equals the duration of liabilities (i.e., the net duration is zero), a financial institution eliminates its exposure to small parallel shifts in the yield curve. It is still exposed to shifts that are either large or nonparallel.

2.9 Convexity

◇ Figure 1.6 shows the relationship between percentage change in bond value and change in its yield for two bond portfolios having the same duration. We can observe a small difference of percentage change in bond value for a small yield change, but a large difference for a large yield change.

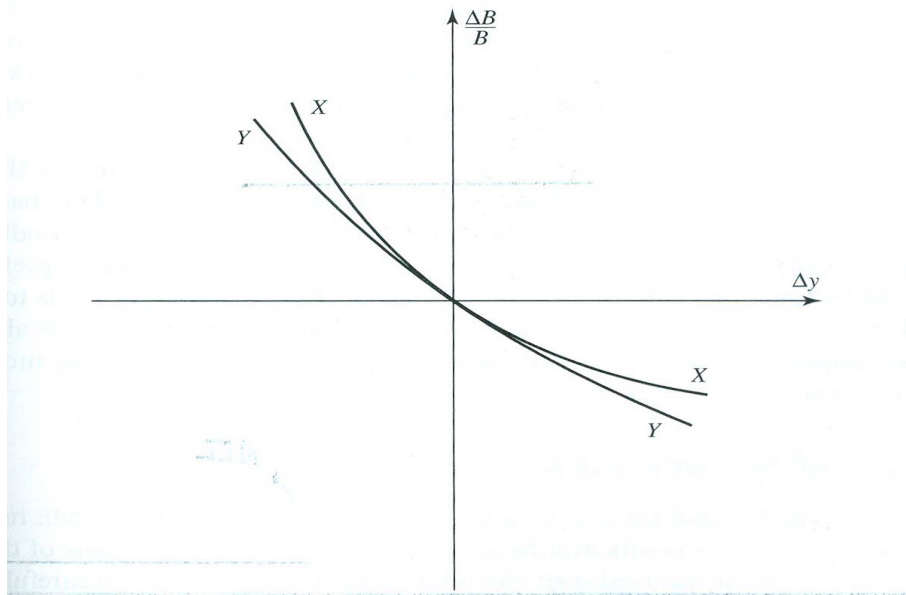


Figure 2.6: Two bond portfolios with the same duration.

◇ The *convexity*, C , of a bond is defined as

$$C = \frac{1}{B} \frac{\partial^2 B}{\partial y^2} = \frac{\sum_{i=1}^n c_i t_i^2 e^{-y t_i}}{B}.$$

This leads to a more accurate relationship of

$$\frac{\Delta B}{B} = -D\Delta y + \frac{1}{2}C \cdot (\Delta y)^2$$

- The convexity of a bond portfolio tends to be greatest when the portfolio provides payments evenly over a long period of time, while it is least when the payments are concentrated around one particular point in time.
 - When used for bond portfolios it allows larger shifts in the yield curve to be considered, but the shifts still have to be parallel.
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